

Chapter Five

Input

Discovering Computers 2012

**Your Interactive Guide
to the Digital World**



Objectives Overview

Define input and differentiate among a program, command, and user response

Identify the keys and buttons commonly found on desktop computer keyboards, and describe how keyboards for mobile computers and devices differ from desktop computer keyboards

Describe different mouse types and explain how to use a mouse

Describe various types of touch screens and explain how a touch-sensitive pad works

Describe various types of pen input, and identify other types of input for smart phones

Summarize the purpose of various game controllers

Objectives Overview

Explain how resolution affects the quality of a picture captured on a digital camera

Describe the uses of voice recognition, Web cams, and video conferencing

Discuss how various scanners and reading devices work

Summarize the various biometric devices

Discuss how POS terminals, automated teller machines, and DVD kiosks work

Identify alternative input devices for physically challenged users

What Is Input?

- **Input** is any data and instructions entered into the memory of a computer



What Is Input?

- Instructions can be entered into the computer in the form of programs, commands, and user responses

A program is a series of related instructions that tells a computer what tasks to perform and how to perform them

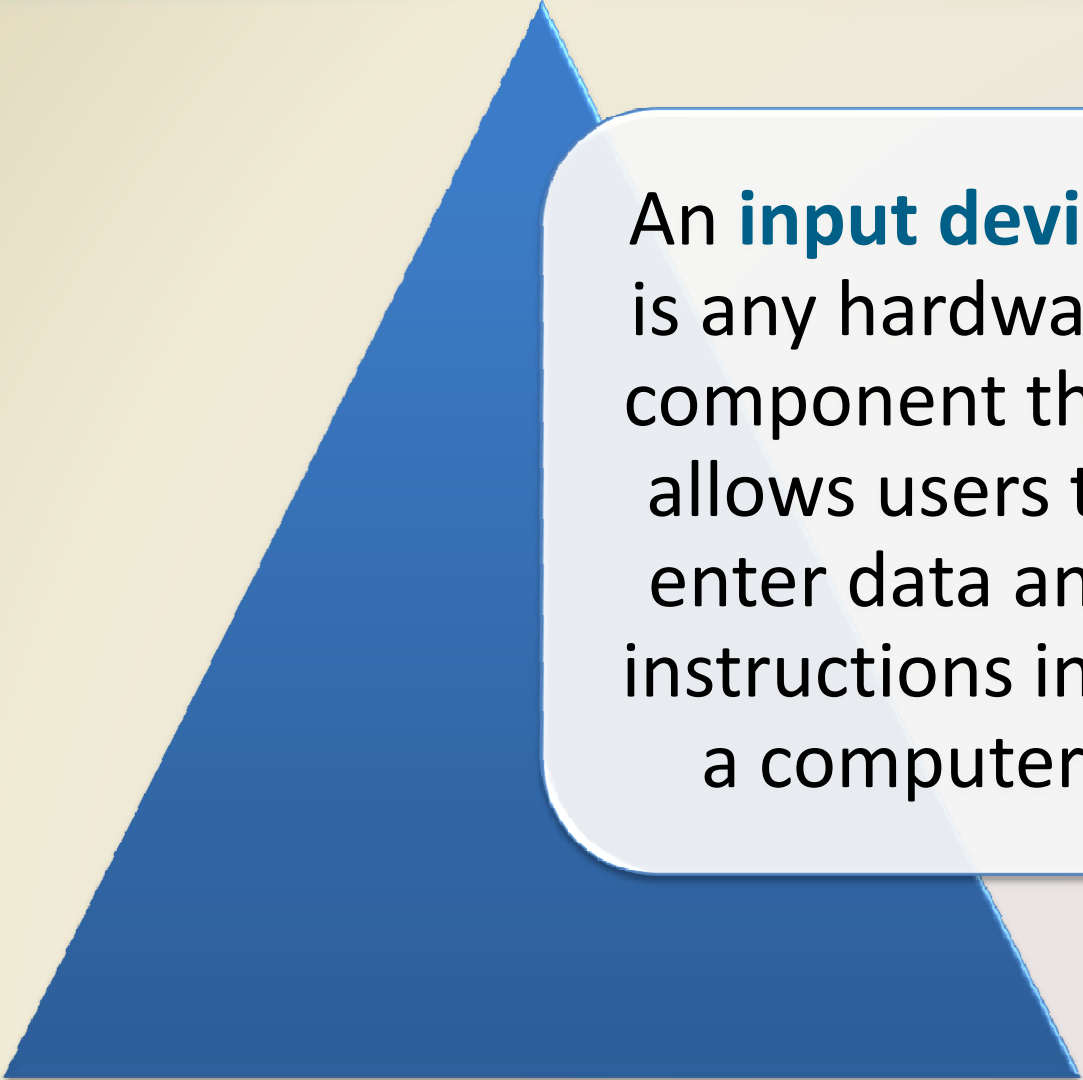


Programs respond to commands that a user issues



A user response is an instruction a user issues by replying to a question displayed by a program

What Are Input Devices



An **input device** is any hardware component that allows users to enter data and instructions into a computer

The Keyboard

- A **keyboard** is an input device that contains keys users press to enter data and instructions into a computer



The Keyboard

- Most desktop computer keyboards have...

Between 101 and
105 keys

A numeric keypad
on the right side of
the keyboard

Function keys,
CTRL keys, ALT
keys, and arrow
keys

WINDOWS key

APPLICATION key

Toggle keys

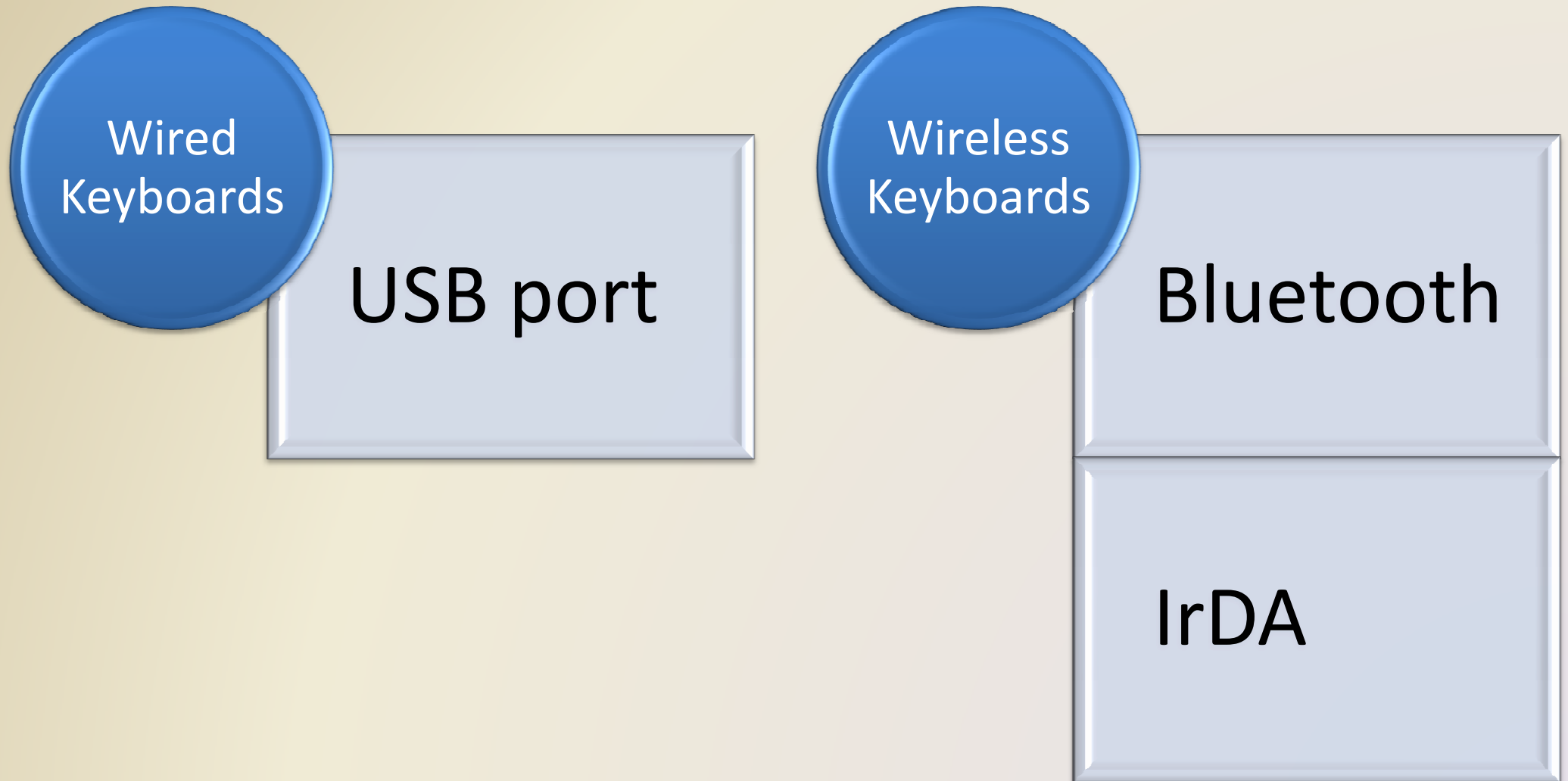


The Keyboard

- The **insertion point**, also known as the cursor, is a symbol on the screen that indicates where the next character you type will appear



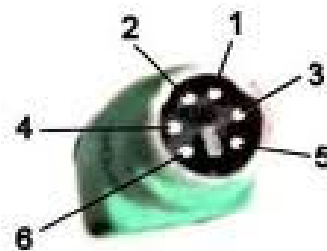
The Keyboard





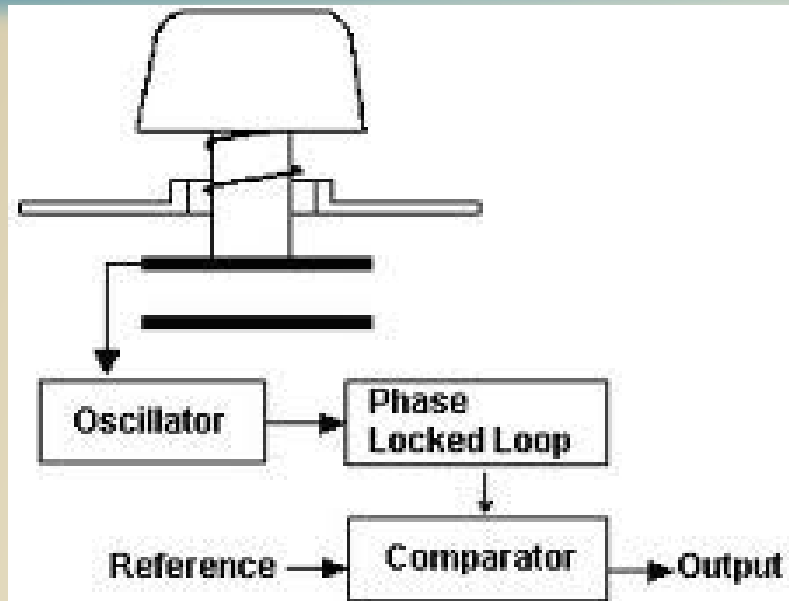
PS/2 keyboard connector (MINI-DIN6)

Connector Pin #	Purpose
Pin 1	KBDAT (data)
Pin 2	not used
Pin 3	GND
Pin 4	VCC (+5V)
Pin 5	KBDCLK (clock)
Pin 6	not used



Mouse connector pinout is identical to PS/2 keyboard.

Keyboard Switch



The Keyboard

- An ergonomic keyboard has a design that reduces the chance of wrist and hand injuries
- **Ergonomics** incorporates comfort, efficiency, and safety into the design of the workplace



The Keyboard

- Keyboards on mobile devices typically are smaller and/or have fewer keys
- Some phones have predictive text input, which saves time when entering text using the phone's keypad



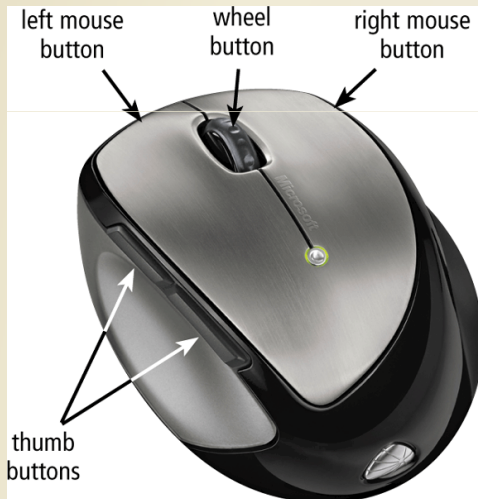
Pointing Devices

A **pointing device** is an input device that allows a user to control a pointer on the screen

A **pointer** is a small symbol on the screen whose location and shape change as a user moves a pointing device

Mouse

- A **mouse** is a pointing device that fits under the palm of your hand comfortably
 - Most widely used pointing device on desktop computers
- A mouse can be wired or wireless



Mouse

- Mouse operations

Point

Click

Right-click

Double-click

Triple-click

Drag

Right-drag

Rotate
wheel

Free-spin
wheel

Press wheel

Tilt wheel

Press thumb
button



Other Pointing Devices



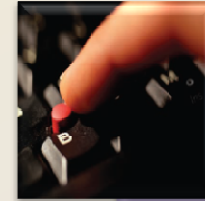
Trackball

- A **trackball** is a stationary pointing device with a ball on its top or side



Touchpad

- A **touchpad** is a small, flat, rectangular pointing device that is sensitive to pressure and motion

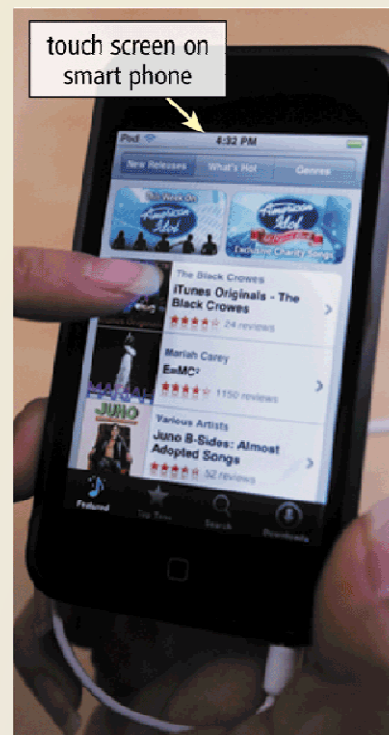
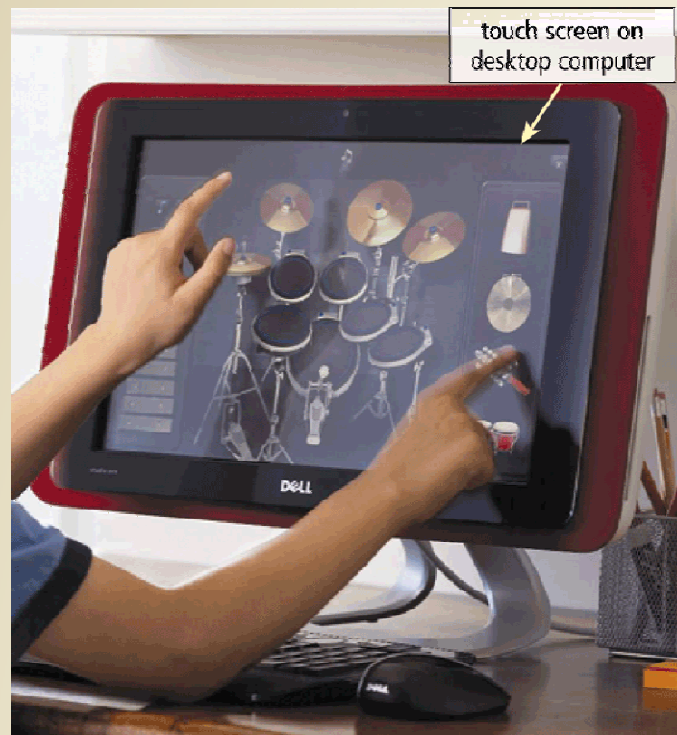


Pointing Stick

- A **pointing stick** is a pressure-sensitive pointing device shaped like a pencil eraser that is positioned between keys on a keyboard

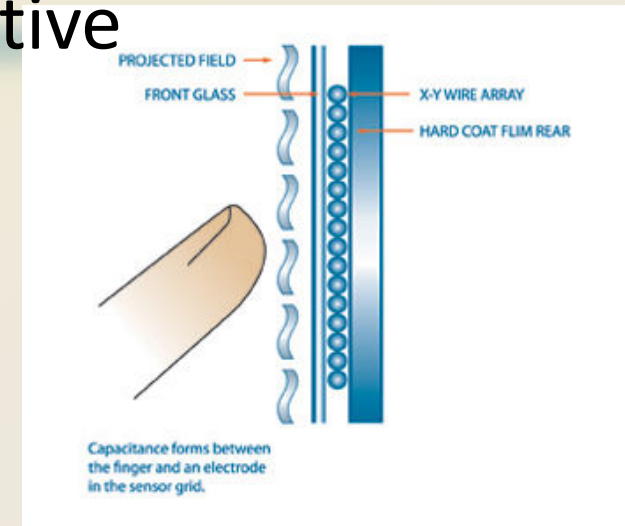
Touch Screens and Touch-Sensitive Pads

- A **touch screen** is a touch-sensitive display device

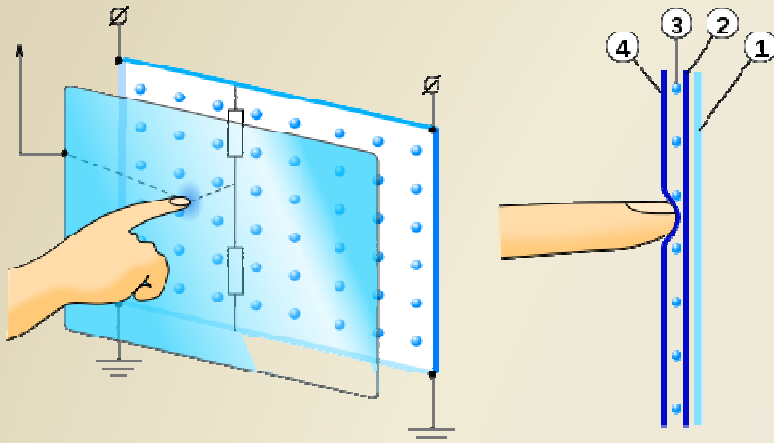


Touch Screen Technologies

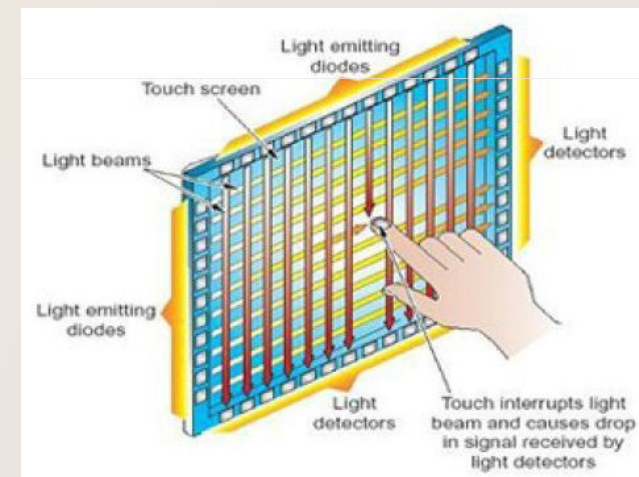
Capacitive



Resistive



Infrared

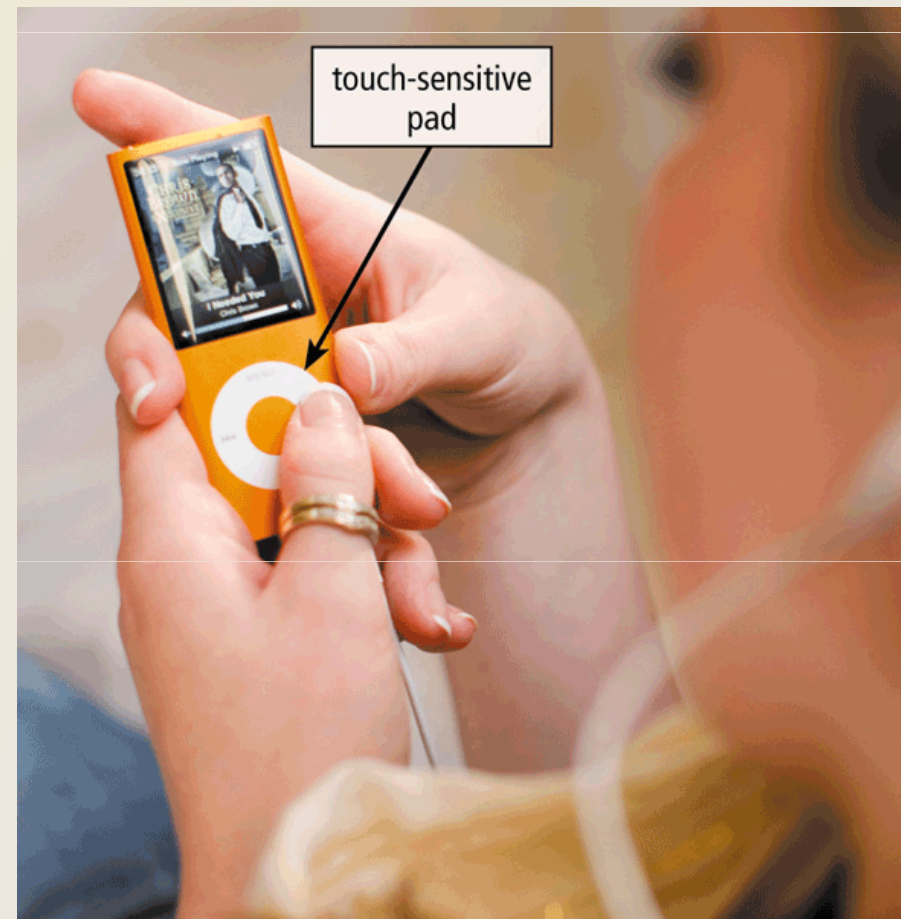


Touch Screens and Touch-Sensitive Pads

Microsoft Surface



Touch-sensitive pads



Pen Input

- With **pen input**, you touch a **stylus** or **digital pen** on a flat surface to write, draw, or make selections



Other Input for Smart Phones



Game Controllers

- Video games and computer games use a **game controller** as the input device that directs movements and actions of on-screen objects

Gamepads

**Joysticks and
Wheels**

Light guns

Dance pads

**Motion-
sensing
controllers**

Game Controllers



Digital Cameras

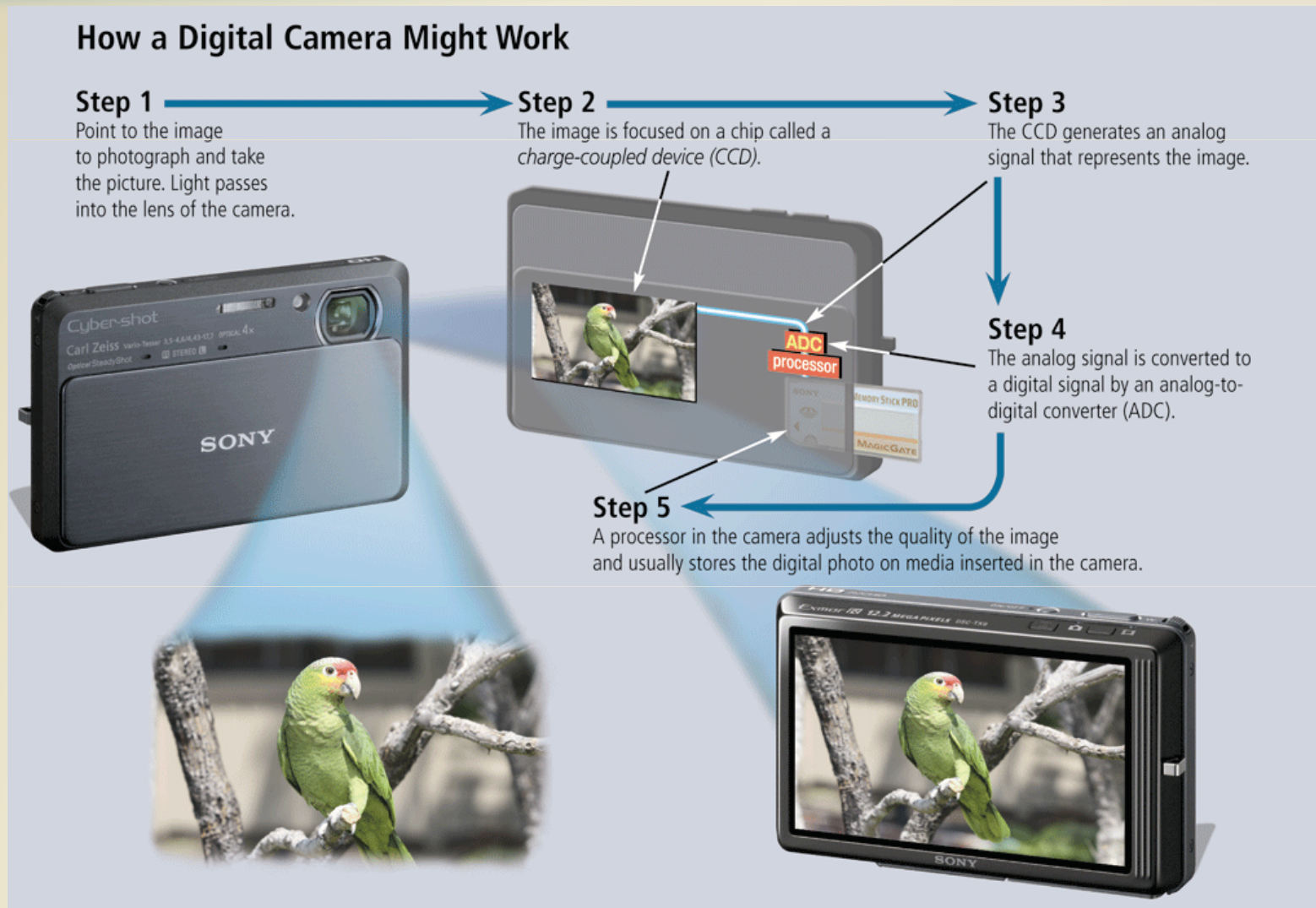
- A **digital camera** is a mobile device that allows users to take pictures and store them digitally

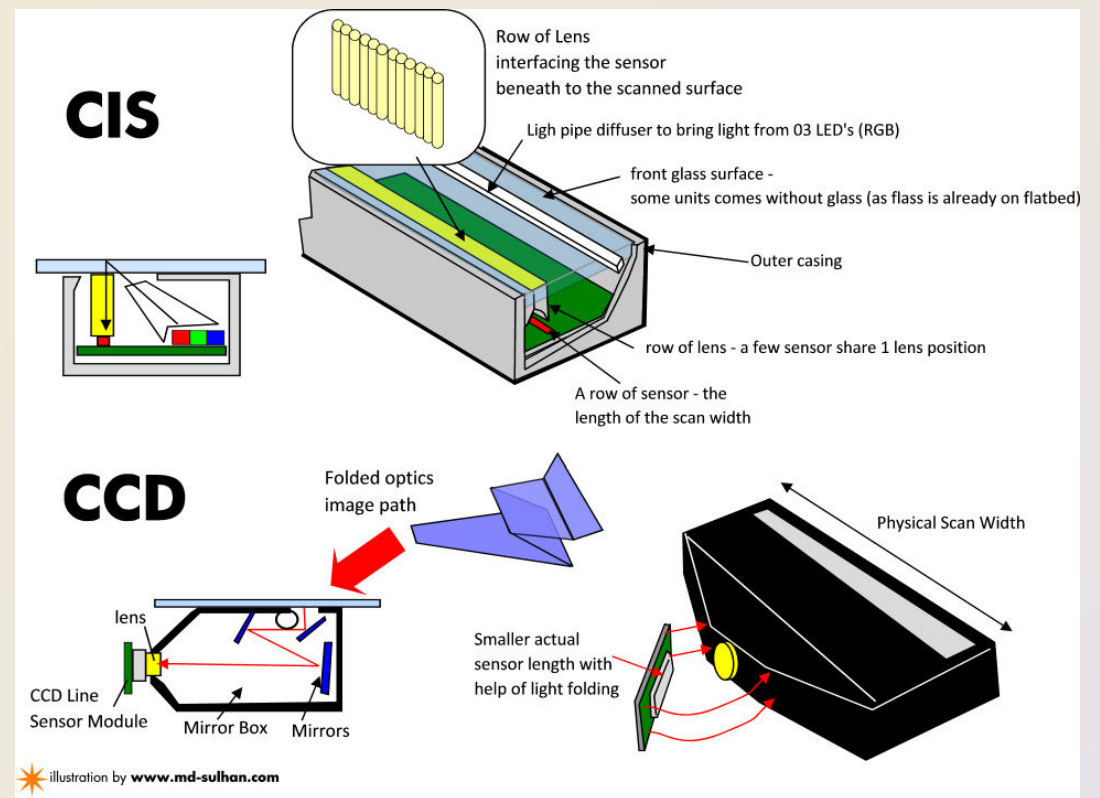
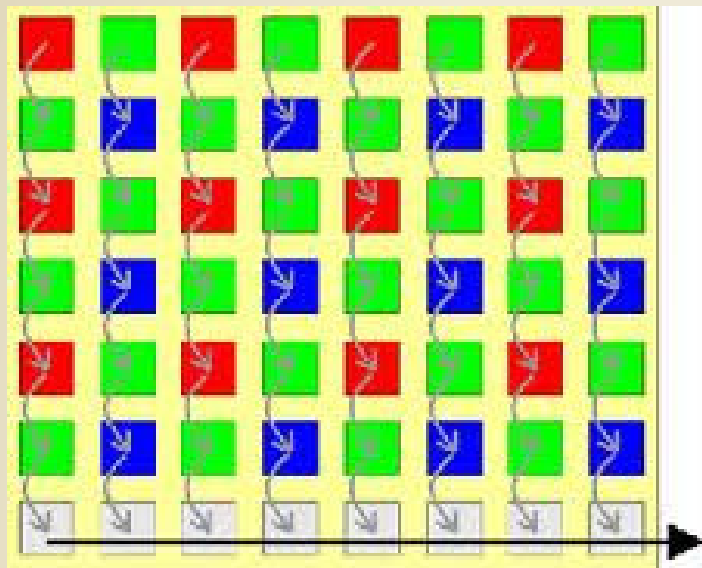
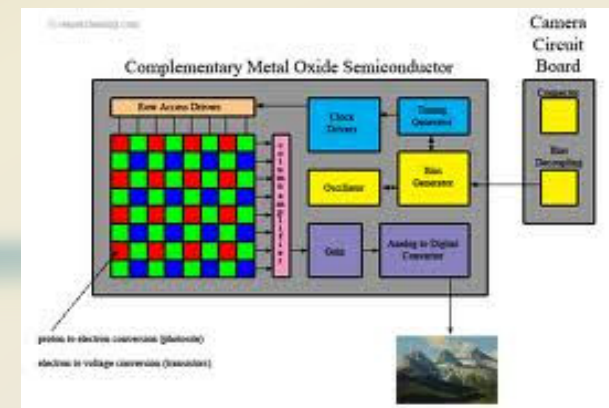
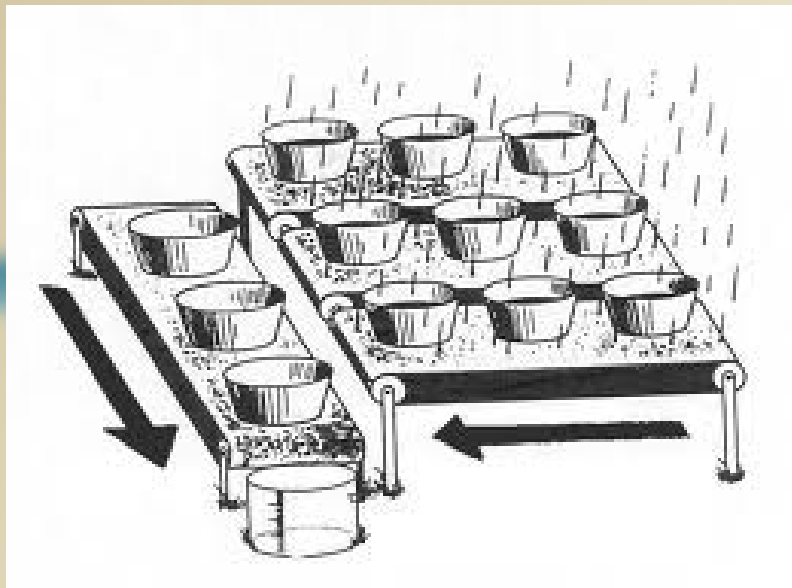
Studio cameras

Field cameras

Point-and-shoot camera

Digital Cameras







Digital Cameras

- Two factors affect the quality of digital camera photos:

Resolution

- **Resolution** is the number of horizontal and vertical pixels in a display device
- A pixel is the smallest element in an electronic display

Number of bits stored in each pixel

- Each pixel consists of one or more bits of data
- The more bits used to represent a pixel, the more colors and shades of gray that can be represented

NEWS TECHNOLOGY

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Video

21 June 2012 Last updated at 13:54 GMT



Gigapixel camera offers super sharp shots

A camera capable of creating images with "unprecedented detail" has been unveiled by US engineers.

The prototype machine - dubbed AWARE2 - has the potential to take pictures with resolutions of up to 50 gigapixels, equivalent to 50,000 megapixels, **according to the team from Duke University in North Carolina.**

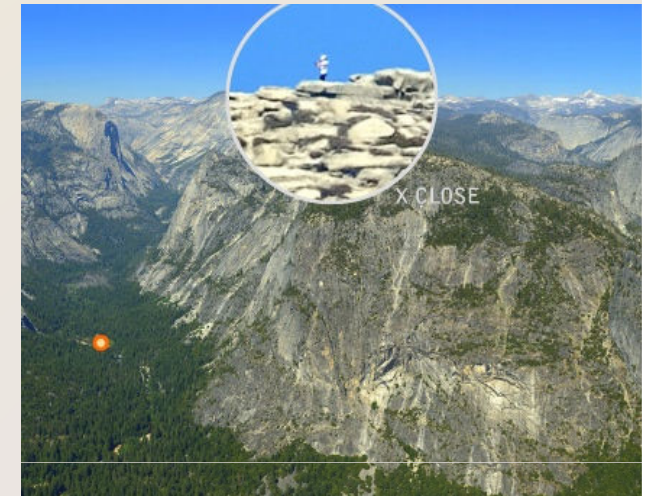
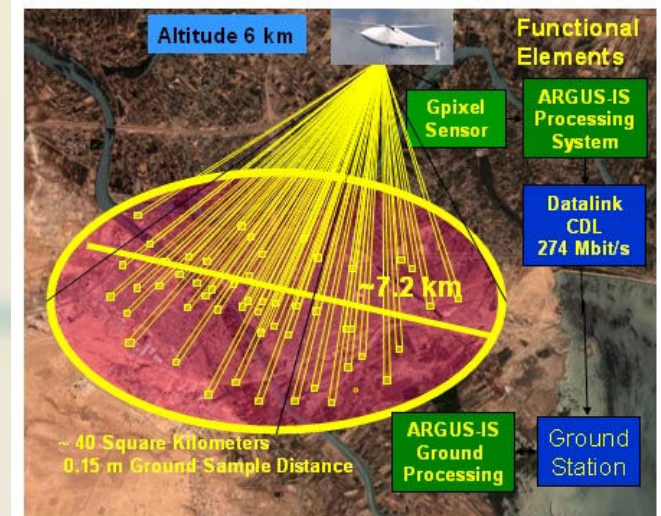
It works by synchronising 98 tiny cameras in a single device.

The machine is likely to be used first for military surveillance.

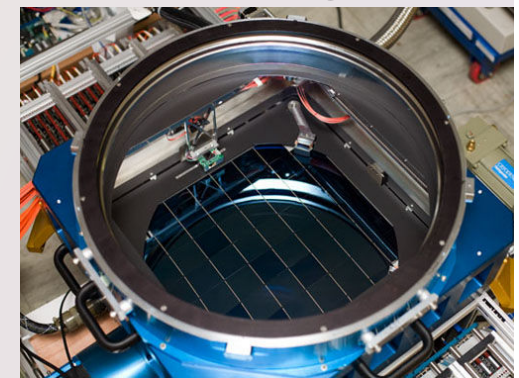
In its current state the researchers say it can take one-gigapixel images at up to three



The current camera is quite bulky



2008, 1.4GigaPixels



Voice Input

- **Voice input** is the process of entering input by speaking into a microphone
- **Voice recognition** is the computer's capability of distinguishing spoken words



Voice Input

- **Audio input** is the process of entering any sound into the computer

Speech

Music

Sound Effects

Voice Input

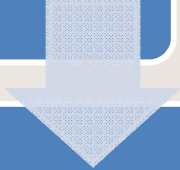
- Music production software allows users to record, compose, mix, and edit music and sounds



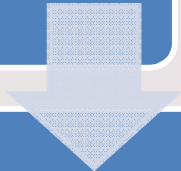
Video Input

- **Video input** is the process of capturing full-motion images and storing them on a computer's storage medium

Record video on a **digital video (DV) camera** or use a video capture card to convert analog signals to digital



Connect the camera to a port on the system unit



Transfer video and images

Video Input



Video: Video Editing on Your Computer



[CLICK TO START](#)

Video Input

- A **Web cam** is a type of digital video camera that enables a user to:

Capture video and still images

Send e-mail messages with video attachments

Add live images to instant messages

Broadcast live images over the Internet

Make video telephone calls

Video Input

- A **video conference** is a meeting between two or more geographically separated people



Scanners and Reading Devices



Flatbed



Pen or Handheld



Sheet-fed



Drum



Scanners and Reading Devices

How a Flatbed Scanner Works

Step 1

Place the document to be scanned face down on the glass window. Using buttons on the scanner or the scanner program, start the scanning process.



Step 2

The scanner converts the document content to digital information, which is transmitted through the cable to the memory of the computer.

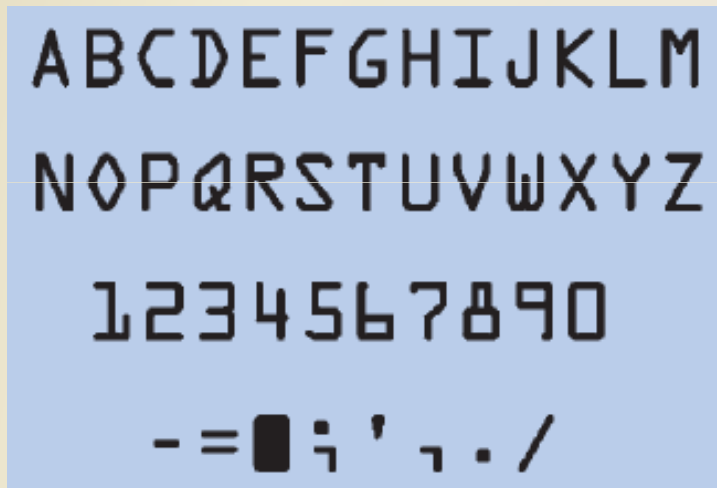
Step 3

Once in the memory of the computer, users can display the image, print it, e-mail it, include it in a document, or place it on a Web page.



Scanners and Reading Devices

- **Optical character recognition (OCR)** involves reading characters from ordinary documents
- A **turnaround document** is a document you return to the company that creates and sends it



CHASE Continental Airlines

ACCOUNT NUMBER: 24085 6105 0824 8058

PAYMENT DUE DATE: 02/14/11

name and address printed using OCR characters

ISAAC NARO
1227 SOUTH HI ST
LOS ANGELES CA 90055

new balance \$1,165.15

minimum due \$23.834

BOX 52064
ITX AZ 85072-2064

408161050924865200116515000023004

OCR characters

Continental Airlines Visa® Signature Card from Chase

ACCOUNT NUMBER: 24085 6105 0824 8058

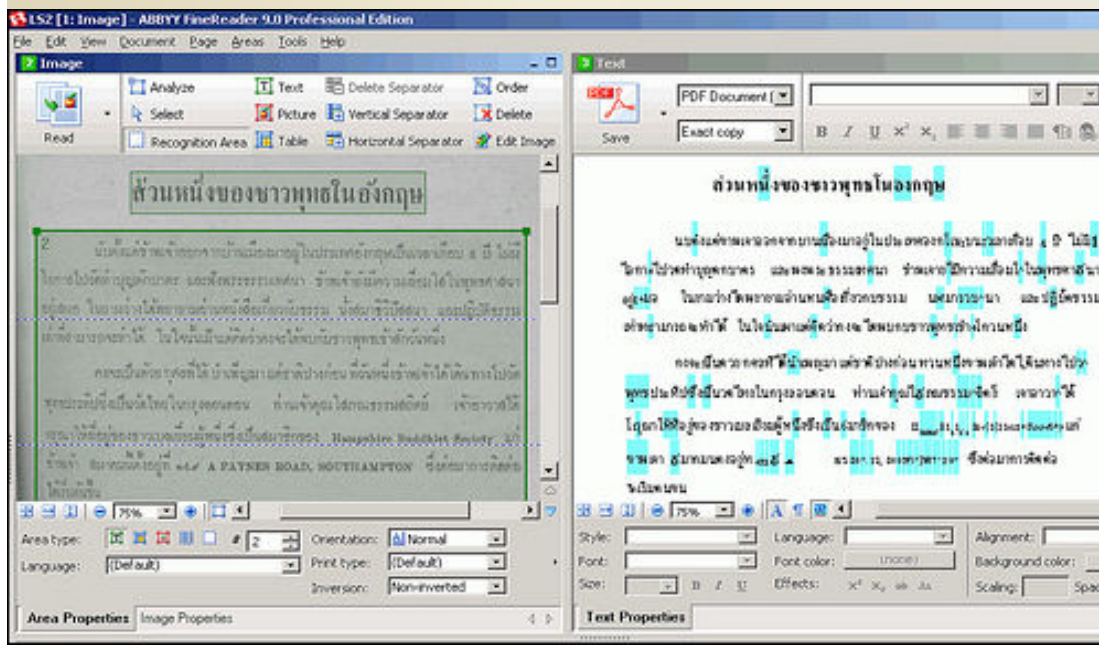
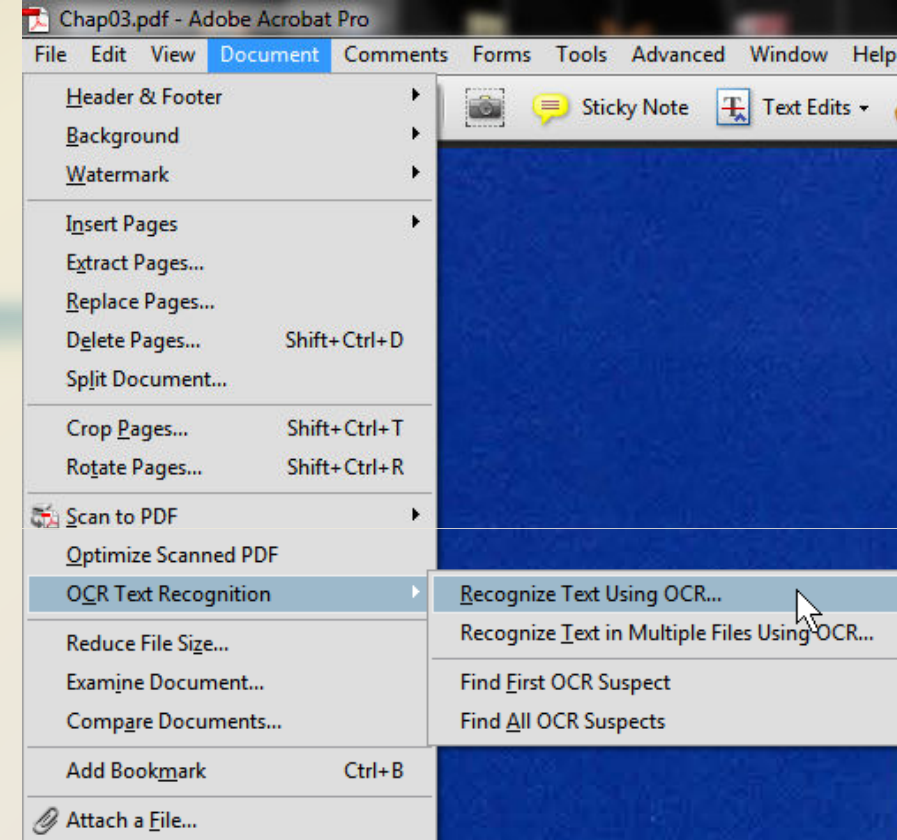
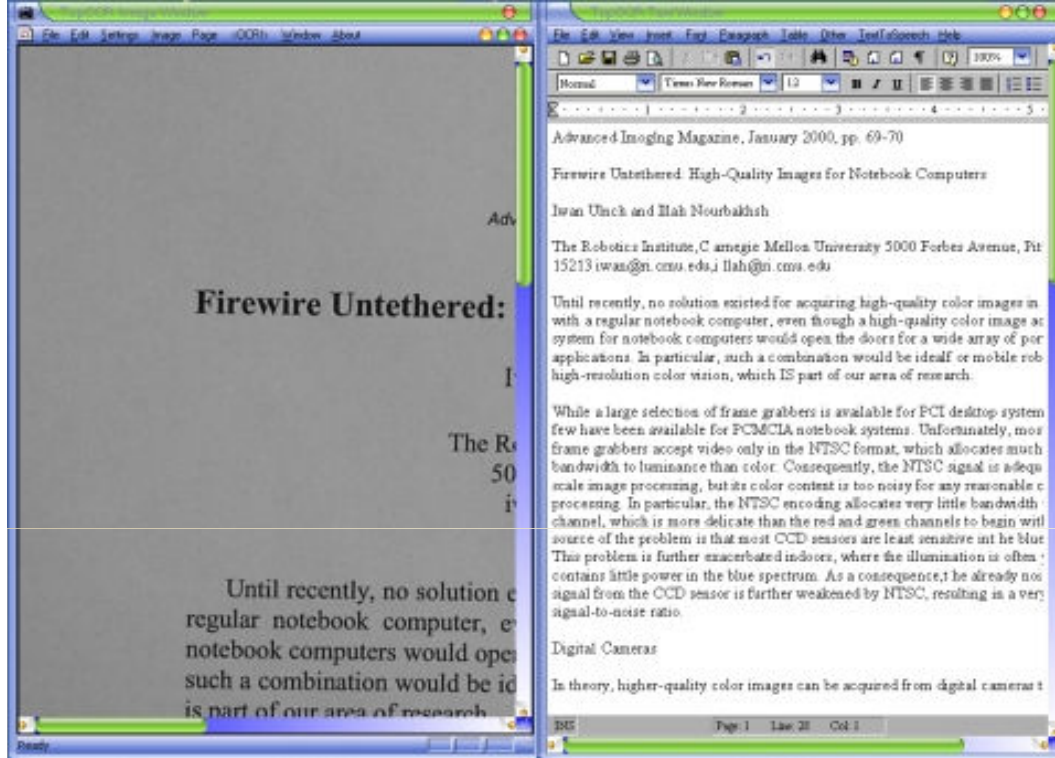
NEW BALANCE		REVOLVING LINE BALANCE		NON REVOLVING LINE BALANCE		STATEMENT CLOSING DATE	
NEW BALANCE	\$1,165.15	REVOLVING LINE BALANCE	\$1,165.15	NON REVOLVING LINE BALANCE	\$0.00	STATEMENT CLOSING DATE	02/18/11
REVOLVING LINE	\$25,000	AVAILABLE REVOLVING LINE	\$23,834	CASH ACCESS LINE	\$25,000	AVAILABLE CASH	\$23,834

ONEPASS® MILEAGE SUMMARY

Miles Earned This Statement	Total Miles Earned Since 01/09
1,014	100,497

Here is your Account Summary:

	AMOUNT
Previous Balance	\$471.50
(+) Payments, Credits	471.50
(-) Purchases, Cash, Debits	1,165.15
(+) FINANCE CHARGES	0.00
(=) New Balance	1,165.15
Total Minimum Payment Due by 02/14/11	\$23.00



Scanners and Reading Devices

- **Optical mark recognition (OMR)** reads hand-drawn marks such as small circles or rectangles
- An OMR device scans the documents and matches the patterns of light





OMR (Optical Mark Reader) Answer Sheet
ALL INDIA OPEN
MATHEMATICS SCHOLARSHIP EXAMINATION
Answer Sheet

EXAMPLE WRONG METHODS 1 ☐ ☒ ☐ ☐ 2 ☐ ☒ ☐ ☐ 3 ☐ ☐ ☒ ☐ 4 ☐ ☒ ☒ ☐		SEAT NO. <table border="1"> <tr> <td>9</td><td>0</td><td>2</td><td>5</td><td>4</td><td>8</td> </tr> <tr> <td>6</td><td>●</td><td>6</td><td>6</td><td>6</td><td>6</td> </tr> <tr> <td>1</td><td>1</td><td>3</td><td>1</td><td>1</td><td>1</td> </tr> <tr> <td>5</td><td>2</td><td>●</td><td>2</td><td>2</td><td>2</td> </tr> <tr> <td>3</td><td>3</td><td>3</td><td>3</td><td>3</td><td>3</td> </tr> <tr> <td>4</td><td>4</td><td>4</td><td>4</td><td>●</td><td>4</td> </tr> <tr> <td>5</td><td>5</td><td>5</td><td>●</td><td>5</td><td>5</td> </tr> <tr> <td>6</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td> </tr> <tr> <td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td> </tr> <tr> <td>8</td><td>8</td><td>8</td><td>8</td><td>8</td><td>●</td> </tr> <tr> <td>●</td><td>9</td><td>9</td><td>9</td><td>9</td><td>9</td> </tr> </table>							9	0	2	5	4	8	6	●	6	6	6	6	1	1	3	1	1	1	5	2	●	2	2	2	3	3	3	3	3	3	4	4	4	4	●	4	5	5	5	●	5	5	6	6	6	6	6	6	7	7	7	7	7	7	8	8	8	8	8	●	●	9	9	9	9	9	NAME <u>Ghosh Saptashwa A.</u> SCHOOL CODE <u>5126</u> STANDARD <u>9th</u> MEDIUM <u>English</u> DATE <u>1st Feb. 2009.</u> PHONE NO. <u>020-25818606</u> MOBILE NO. <u>9822441139</u>	
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CORRECT METHODS 1 ☐ ☐ ☒ ☐		CANDIDATE'S SIGNATURE <u>Al Jia</u> SUPERVISOR'S SIGNATURE _____																																																																										

INSTRUCTIONS

1. There is only one correct answer for each question
2. All entries in the circle must be made by BLUE or BLACK Ink Ball Pen only. Do not try to alter the entry.
3. Circle should be darkened completely so that the numeral inside the circle is not visible.
4. Do not use any stray marks on the sheet for rough work use sheet provided separately for it.
5. Mark your answer as shown in the example.

1	☐ 1 ☒ 2 ☐ 3 ☐ 4	11	☐ 1 ☐ 2 ☐ 3 ☐ 4	21	☐ 1 ☐ 2 ☐ 3 ☐ 4	31	☐ 1 ☐ 2 ☐ 3 ☐ 4	41	☐ 1 ☐ 2 ☐ 3 ☐ 4
2	☐ 1 ☐ 2 ☐ 3 ☒ 4	12	☐ 1 ☐ 2 ☐ 3 ☐ 4	22	☐ 1 ☐ 2 ☐ 3 ☐ 4	32	☐ 1 ☐ 2 ☐ 3 ☐ 4	42	☐ 1 ☐ 2 ☐ 3 ☐ 4
3	☐ 1 ☐ 2 ☒ 3 ☐ 4	13	☐ 1 ☐ 2 ☐ 3 ☐ 4	23	☐ 1 ☐ 2 ☐ 3 ☐ 4	33	☐ 1 ☐ 2 ☐ 3 ☐ 4	43	☐ 1 ☐ 2 ☐ 3 ☐ 4
4	☐ 1 ☒ 2 ☐ 3 ☐ 4	14	☐ 1 ☐ 2 ☐ 3 ☐ 4	24	☐ 1 ☐ 2 ☐ 3 ☐ 4	34	☐ 1 ☐ 2 ☐ 3 ☐ 4	44	☐ 1 ☐ 2 ☐ 3 ☐ 4
5	☒ 1 ☐ 2 ☐ 3 ☐ 4	15	☐ 1 ☐ 2 ☐ 3 ☐ 4	25	☐ 1 ☐ 2 ☐ 3 ☐ 4	35	☐ 1 ☐ 2 ☐ 3 ☐ 4	45	☐ 1 ☐ 2 ☐ 3 ☐ 4
6	☐ 1 ☐ 2 ☐ 3 ☐ 4	16	☐ 1 ☐ 2 ☐ 3 ☐ 4	26	☐ 1 ☐ 2 ☐ 3 ☐ 4	36	☐ 1 ☐ 2 ☐ 3 ☐ 4	46	☐ 1 ☐ 2 ☐ 3 ☐ 4
7	☐ 1 ☐ 2 ☐ 3 ☐ 4	17	☐ 1 ☐ 2 ☐ 3 ☐ 4	27	☐ 1 ☐ 2 ☐ 3 ☐ 4	37	☐ 1 ☐ 2 ☐ 3 ☐ 4	47	☐ 1 ☐ 2 ☐ 3 ☐ 4
8	☐ 1 ☐ 2 ☐ 3 ☐ 4	18	☐ 1 ☐ 2 ☐ 3 ☐ 4	28	☐ 1 ☐ 2 ☐ 3 ☐ 4	38	☐ 1 ☐ 2 ☐ 3 ☐ 4	48	☐ 1 ☐ 2 ☐ 3 ☐ 4
9	☐ 1 ☐ 2 ☐ 3 ☐ 4	19	☐ 1 ☐ 2 ☐ 3 ☐ 4	29	☐ 1 ☐ 2 ☐ 3 ☐ 4	39	☐ 1 ☐ 2 ☐ 3 ☐ 4	49	☐ 1 ☐ 2 ☐ 3 ☐ 4
10	☐ 1 ☐ 2 ☐ 3 ☐ 4	20	☐ 1 ☐ 2 ☐ 3 ☐ 4	30	☐ 1 ☐ 2 ☐ 3 ☐ 4	40	☐ 1 ☐ 2 ☐ 3 ☐ 4	50	☐ 1 ☐ 2 ☐ 3 ☐ 4

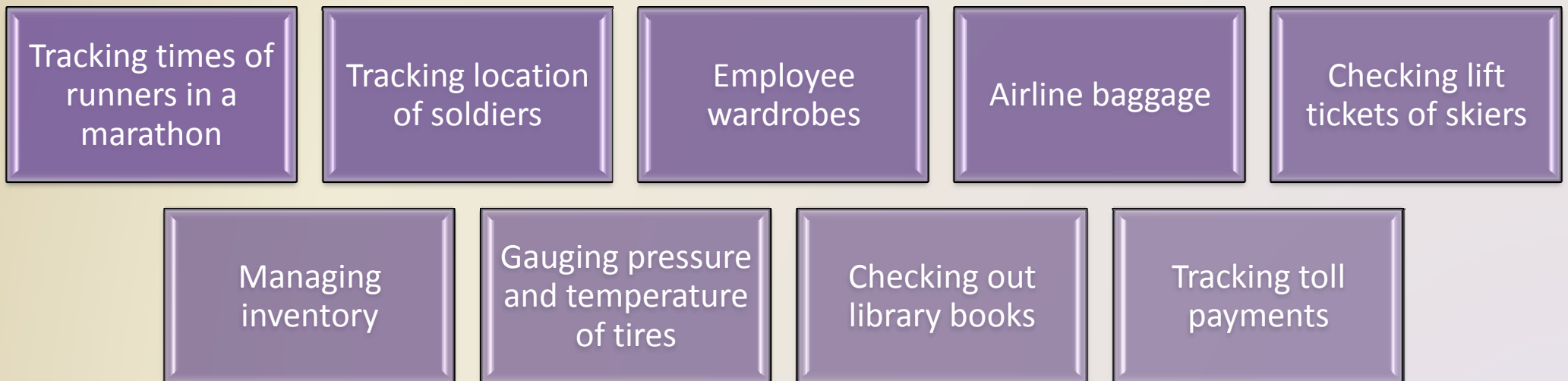
Scanners and Reading Devices

- A **bar code reader**, also called a **bar code scanner** uses laser beams to read **bar codes**



Scanners and Reading Devices

- **RFID** (radio frequency identification) uses radio signals to communicate with a tag placed in or attached to an object
- An **RFID reader** reads information on the tag via radio waves
- RFID can track:



Scanners and Reading Devices

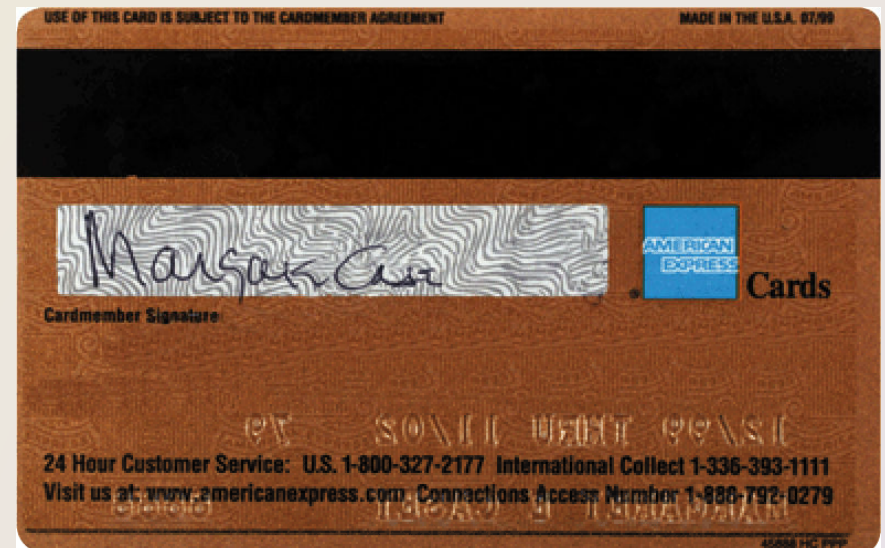
- **Magnetic stripe card readers** read the magnetic stripe on the back of cards such as:

Credit cards

Entertainment cards

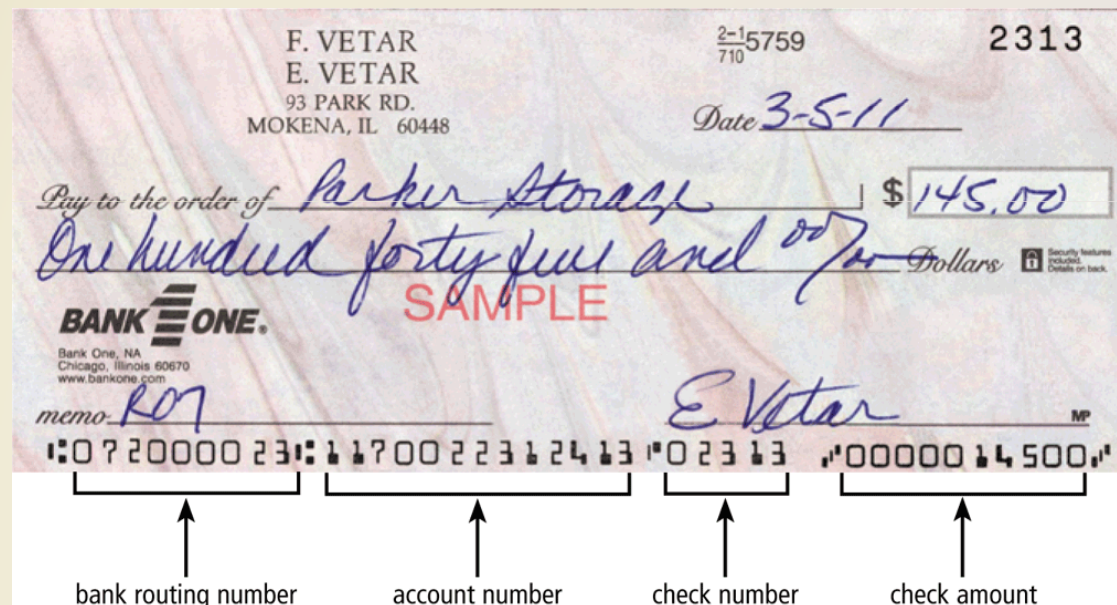
Bank cards

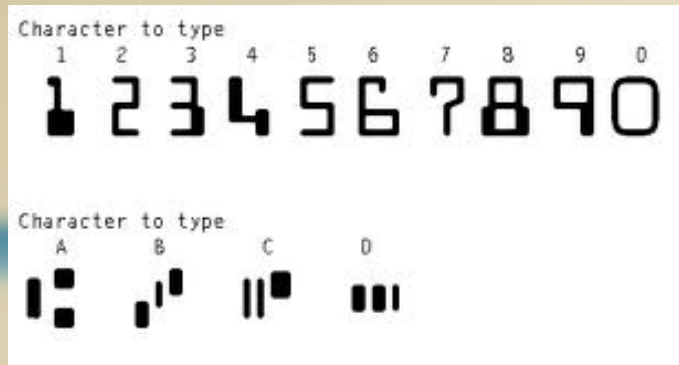
Other similar cards



Scanners and Reading Devices

- **MICR** (magnetic ink character recognition) devices read text printed with magnetized ink
- An **MICR reader** converts MICR characters into a form the computer can process
- Banking industry uses MICR for check processing





Scanners and Reading Devices

- Data collection devices obtain data directly at the location where the transaction or event takes place
- Used in:
 - Restaurants
 - Grocery stores
 - Factories
 - Warehouses
 - The outdoors



Biometric Input

- Biometrics authenticates a person's identity by verifying a personal characteristic

**Fingerprint
reader**

Face
recognition
system

Hand
geometry
system

Voice
verification
system

Signature
verification
system

Iris recognition
system

Retinal
scanners

Biometric Input



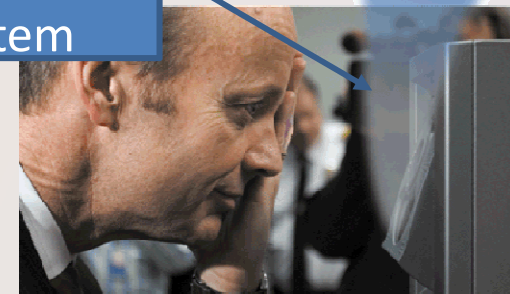
fingerprint
reader



hand
geometry
system



iris
recognition
system



Terminals

- A terminal is a computer that allows users to send data to and/or receive information from a host computer



A **POS terminal** records purchases, processes payment, and updates inventory

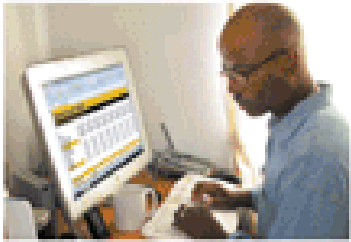


An **automated teller machine (ATM)** allows users to access their bank accounts



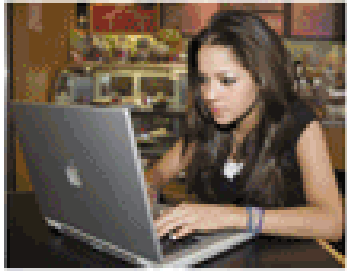
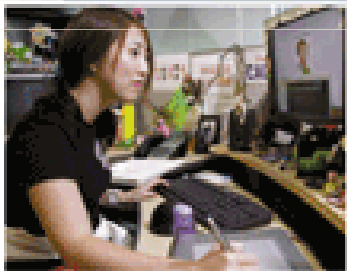
A **DVD kiosk** is a self-service DVD rental machine

Putting It All Together

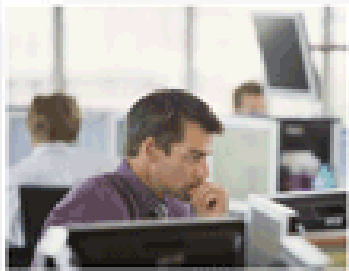
Suggested Input Devices by User	
User	Input Device
Home 	• Enhanced keyboard or ergonomic keyboard • Mouse • Stylus for smart phone or other mobile device • Game controller(s) • 30-bit 600 × 1200 ppi color scanner • 7 MP digital camera • Headphones that include a microphone (headset) • Web cam • Fingerprint reader
Small Office/ Home Office 	• Enhanced keyboard or ergonomic keyboard • Mouse • Stylus and portable keyboard for smart phone or other mobile device, or digital pen for Tablet PC • 36-bit 600 × 1200 ppi color scanner • 8 MP digital camera • Headphones that include a microphone (headset) • Web cam

Putting It All Together

Suggested Input Devices by User

User	Input Device
Mobile 	• Wireless mouse for notebook computer • Touchpad or pointing stick on notebook computer • Stylus and portable keyboard for smart phone or other mobile device, or digital pen for Tablet PC • 7 or 8 MP digital camera • Headphones that include a microphone (headset) • Fingerprint reader for notebook computer
Power 	• Enhanced keyboard or ergonomic keyboard • Mouse • Stylus and portable keyboard for smart phone or other mobile device • Pen for graphics tablet • 48-bit 1200 × 1200 ppi color scanner • 9 to 12 MP digital camera • Headphones that include a microphone (headset) • Web cam

Putting It All Together

Suggested Input Devices by User	
User	Input Device
Enterprise 	• Enhanced keyboard or ergonomic keyboard • Mouse • Stylus and portable keyboard for smart phone or other mobile device, or digital pen for Tablet PC • Touch screen • 42-bit 1200 × 1200 ppi color scanner • 9 to 12 MP digital camera • OCR/OMR readers, bar code readers, MICR reader, or data collection devices • Microphone • Video camera for video conferences • Fingerprint reader or other biometric device

Input Devices for Physically Challenged Users

- Several input devices are available to assist physically challenged users:

Keyguard

Keyboards
with larger
keys

On-screen
keyboard

Various
pointing
devices

Head-
mounted
pointer

Gesture
recognition

Computerized
implant
devices

Input Devices for Physically Challenged Users

Keyboard with larger keys



Head-mounted pointer



Summary

Various techniques of entering input

Several commonly used input devices

Keyboard, mouse, and other pointing devices; touch screens, pen input, other input for smart phones, game controllers, digital cameras, voice input, video input, scanners and reading devices, biometric input, and terminals

Input devices for physically challenged users

Chapter Five

Input

Discovering Computers 2012

**Your Interactive Guide
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Chapter 5 Complete

